

Bernstein Energy & Power: Taking it to the NaX. Are sodium-ion batteries the key to solar baseload power?

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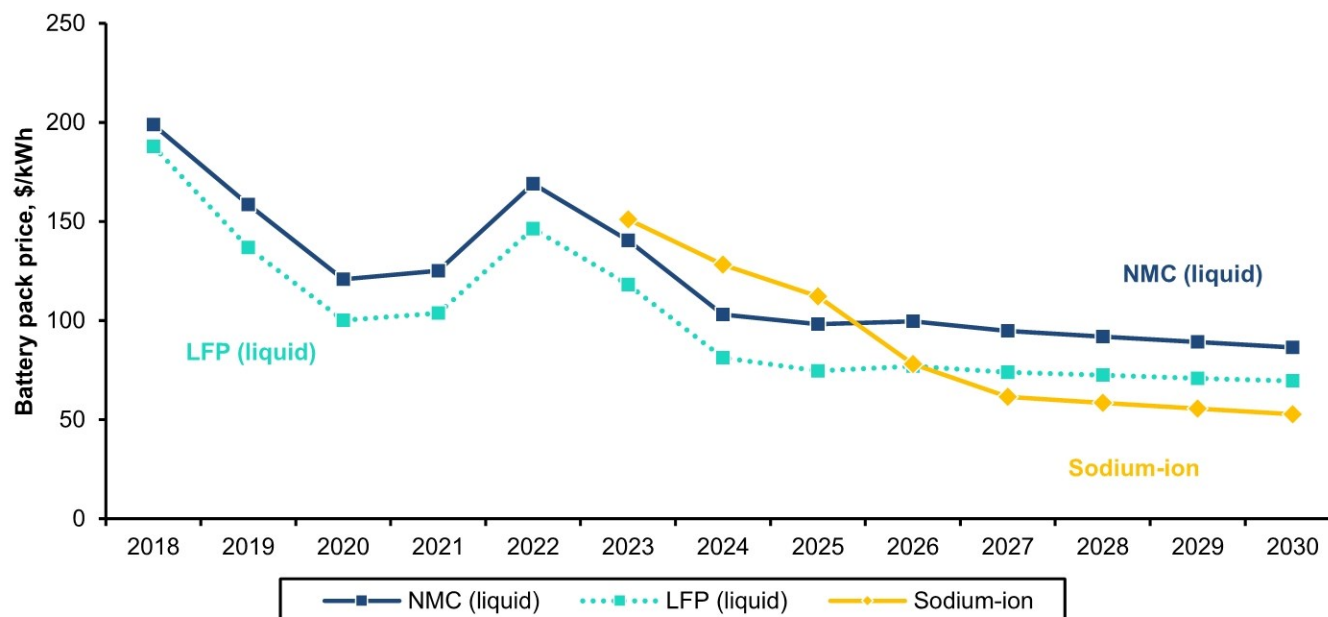
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Taking it to the NaX. Are sodium-ion batteries the key to solar base load power?

Our recent travels in Europe have highlighted two things. Firstly, how hot and sunny it is. In Germany this month roughly 25% of power generation came from solar while in Spain solar share of power generation reached as high as 70% on June 6th. Secondly, client interest in energy storage and particularly sodium-ion batteries has been eye-opening. Many will be aware sodium-ion batteries are the latest technology breakthrough in the battery market. As Elon Musk likes to point out, making batteries out of sodium and iron which are two of the most abundant elements in

the earth's crust (unlike lithium and cobalt), seems like a smart thing to do. The challenge up to now is that making a sodium-ion battery has been expensive. Not because of the cost of the materials, but because of the complexity of the cathode and anode technology. This changed last year with a rapid reduction in the cost of sodium-ion cathodes. While sodium-ion batteries are still as expensive as LFP, all indications suggest they will get cheaper as battery makers scale up production. Robin Zeng, the Chairman of CATL, believes that the cost of sodium-ion batteries will fall to US\$50/kWh, which is lower than the US\$80-90/kWh for LFP and almost half the cost of NMC batteries at over US\$100/kWh.

EXHIBIT 1: Sodium-ion battery prices are expected to match LFP in 2026 and gain further cost advantage thereafter



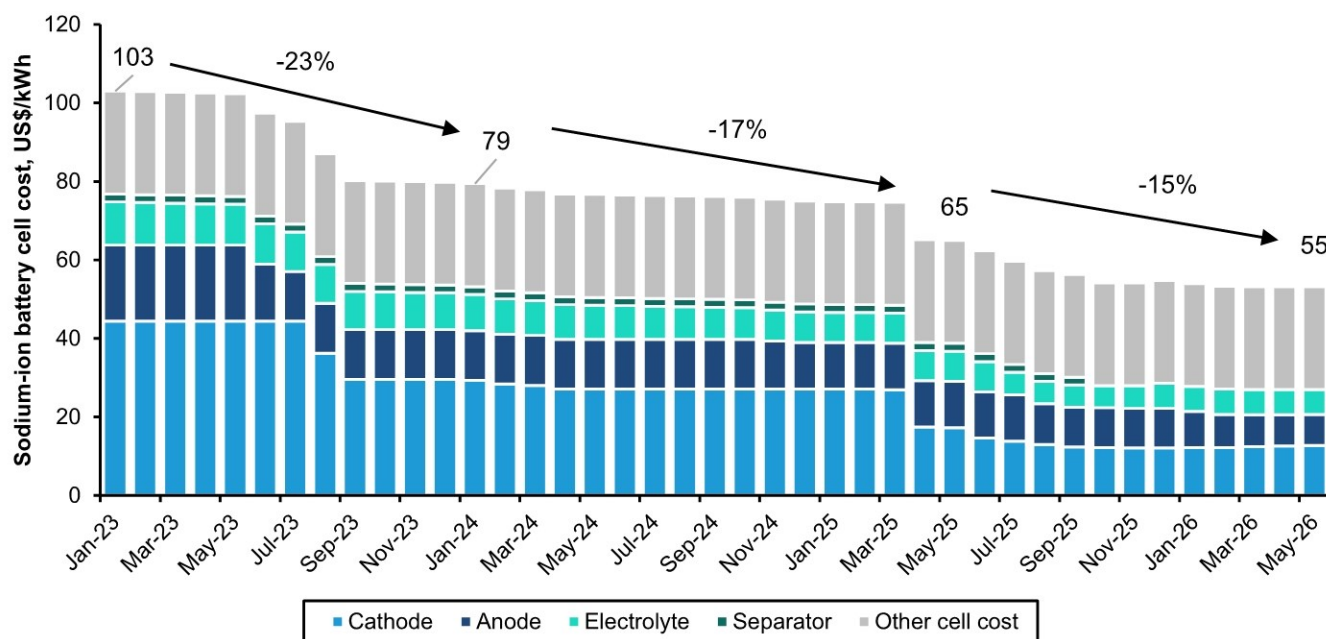
Estimated for 2026-2030

Source: Bloomberg, Baiinfo, Bernstein analysis and estimates

Given that industrial grade sodium costs around US\$5000/t compared to lithium carbonate at over US\$20,000/t, the cost savings are obvious. But the advantages of sodium-ion batteries go beyond just cost. Firstly, the number of cycles that can be achieved with sodium-ion batteries are twice that of LFP. While an LFP battery may be capable of 6,000-8,000 cycles, a Na-ion battery can achieve 15,000 cycles. This means it is possible to charge and discharge a battery twice a day for 20 years, allowing them to be paired directly with solar and wind over the life of the project with no replacement needed. It also means that on a US \$/kWh/cycle basis, Na-ion could be one third the cost of LFP. Secondly, Na-ion batteries are lower temperature, which means

that safety performance is better with thermal runaway limited to temperatures of around 200deg C. Thirdly, Na-ion batteries have better performance at low temperatures. According to CATL, energy retention is greater than 92% at temperatures as low as -20deg C. Finally, while Na has a lower energy density than Li, the overall energy density of Na-ion batteries is getting close to LFP. The CATL Naxtra battery has an energy density of 185Wh/kg at the pack level, which approaches that of LFP (190-210Wh/kg) making Na-ion batteries a serious challenger in the more price-sensitive mass vehicle market.

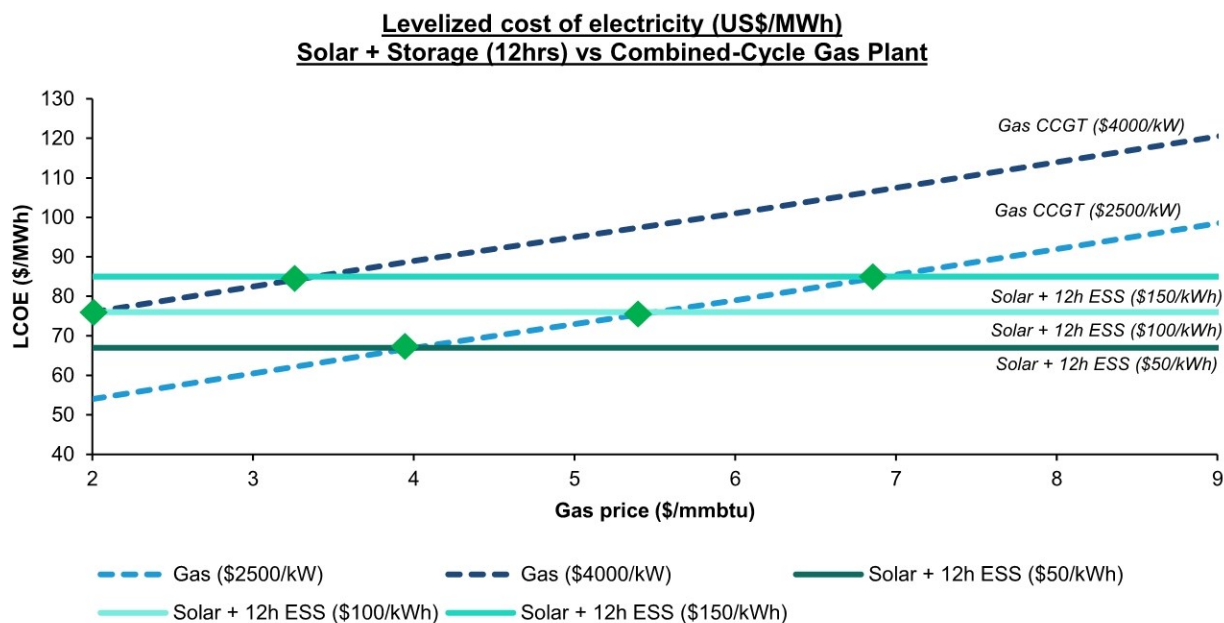
EXHIBIT 2: Based on the spot battery component prices, the cost of sodium-ion battery declined 15% yoy from US\$66/kWh to US\$55/kWh now



For polyanion-based sodium-ion batteries
Source: Baiinfo, Bernstein analysis

The key benefit which Na-ion batteries bring is lower cost. We believe that this can significantly enhance the penetration of renewable energy and substantially increase the total addressable market for batteries. As everyone knows, solar and wind power is cheap, but not when the sun doesn't shine or the wind doesn't blow. The only way to make renewable power cheaper and more effective is to reduce energy storage costs. While lithium-ion batteries are the obvious way to do this, the problem is the high cost of storage. In 2020, it was not uncommon for a BESS to cost US \$500/kWh. This means that 1kW of capacity with 4 hour storage would cost US\$2000, which is almost the same cost as a combined cycle gas turbine plant (US\$2000/kW). Today, the cost of ESS has fallen along with LFP battery costs to around US\$150/kW (battery plus inverter and balance of plant). At the same time, given the demand growth for gas turbines, the capital cost of a gas-fired power plant has increased to as high as US\$4000/kW. Assuming

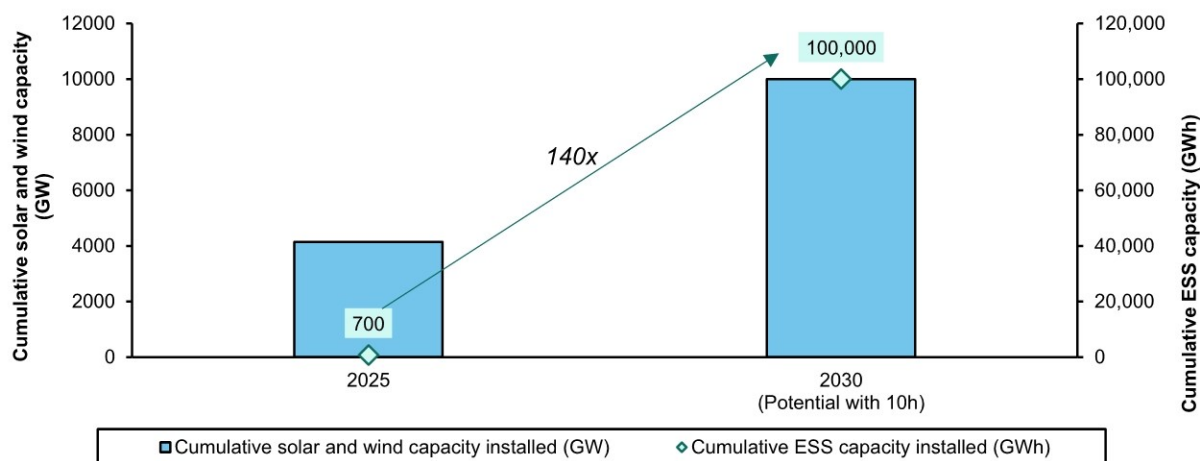
US\$W1000/kW for a solar module, this means that solar plus 20 hours of battery storage (20 x US\$150/kW = US\$3000/kW) could cost as much as a combined cycle gas fired power plant CCGT (US\$4000/kW). But assuming that the cost of Na-ion batteries falls to as low as US\$50/KW, then solar plus battery storage could be even cheaper. It could be possible in the near future that the cost of solar plus 48 hours of Na-ion storage could be as cheap as a gas-fired power plant. On our analysis, the break-even gas price for solar (US\$550/kW) paired with a Na-ion ESS system (US\$100/KWh) and a CCGT US\$2,500/kW could be as low as US\$5/mscf. At a CCGT with a cost of US\$4000/kW the breakeven could be as low as US\$2/mscf. This is not to say that 12 hours of storage would suffice in every location, but the cheaper battery storage becomes the higher the penetration could be.

EXHIBIT 3: Solar with Na-ion storage could compete with CCGT at \$2-5/mmcf gas price

Source: Bernstein analysis

Already in the UAE, Masdar is developing the world first baseload power project with solar and storage. In the project, solar will be paired with 19GWh of battery storage to provide continuous base load solar power for a 1GW of power with a 99.7% power reliability. As battery storage costs come down, longer duration solar plus battery projects could emerge as more serious competitors to gas

for baseload power. This could not only expand the penetration of solar and wind in the grid, but it will help lower power prices and could lead to a significant expansion of battery storage. As of last year there was almost 5,000GW of combined installed solar and wind capacity worldwide.

EXHIBIT 4: Assuming every GW of solar and wind capacity is backed up by 1GW of ESS with 10 hours duration, this could amount to a cumulative demand of 100,000GWh

Source: Bloomberg, Bernstein estimates and analysis

By 2030, we think this number could reach close to 10,000GW. Assuming every GW of solar and wind capacity is backed up by 1GW of battery capacity with a duration of 10 hours this could

amount to a cumulative demand of 100,000GWh. This is two orders of magnitude more than the 700GWh of cumulative installed battery capacity that exists today and would imply significant

upside to the 1000GWh of annual capacity additions we expect this year.

The key investment implication is that Na-ion batteries could enable a step change reduction in energy storage costs allowing solar and wind to not only increase their penetration in the grid but also

to compete more effectively with gas for baseload power supply. This is positive for renewables supply chains but in particular CATL which is the industry leader in Na-ion battery technology and with their recently announced TENER sodium-ion storage system, likely to be an early winner.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	24 Jun 2026		TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
			Closing Price	Price Target		Cur	2025A	2026E	2027E	2025A	2026E	2027E
300750.CH (CATL)	O	CNY	395.36	800.00	17.5%	CNY	16.14	21.95	28.77	24.5	18.0	13.7
3750.HK (CATL)	M	HKD	721.50	770.00	88.8%	CNY	16.14	21.95	28.77	38.8	28.6	21.8
051910.KS (LG Chem)	M	KRW	298,000	298,000	0.9%	KRW (13,258.70)	3,030.56	27,968		(22.5)	98.3	10.7
373220.KS (LGES)	M	KRW	366,000	347,000	(16.5)%	KRW (5,308.10)	1,811.04	9,452.97		(69.0)	202.1	38.7
006400.KS (SDI)	M	KRW	490,500	520,000	140.3%	KRW (9,933.80)	2,099.51	18,376		(49.4)	233.6	26.7
247540.KS (EcoPro BM)	U	KRW	151,800	140,000	4.7%	KRW	403.00	854.00	1,963.00	376.7	177.8	77.3
300274.CH (Sungrow)	O	RMB	154.80	185.00	96.0%	RMB	6.55	8.22	9.33	23.6	18.8	16.6
002466.CH (Tianqi Lithium)	O	CNY	66.13	80.00	70.5%	CNY	0.28	3.90	7.75	234.3	16.9	8.5
9696.HK (Tianqi Lithium)	O	HKD	43.32	65.00	18.2%	CNY	0.28	3.90	7.75	133.3	9.6	4.9
ASIA			1,996.75									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

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Underperform	SELL	12.6%	14.9%

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